

Ethan Shih

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EDUCATION

University of California, Irvine

2025 – present

Computer Science and Engineering, B.S.

Irvine, CA

- Coursework: ICS H32 Python Programming and Libraries (Accelerated), ICS 33 Intermediate Programming, EECS 31 Digital Systems, ICS 45C C++ Programming, EECS 70A Network Analysis

EXPERIENCE

Legacy Robotics

Software Subteam Member

2026 – present

- Developed a PyQt-based mission control GUI for a University Rover Challenge rover, with 4 mission-specific dashboard tabs each surfacing telemetry, rover status, sensor data, controls, and GNSS navigation, alongside a dedicated camera array window.
- Designed the GUI around interchangeable data sources, enabling simulation data for dashboard development and testing ahead of ROS2 integration.
- Collaborated with software and systems leads and a 25-person cross-disciplinary rover team to define dashboard requirements and plan ROS2 interfaces for telemetry, controls, and mission data.

ClassChat

Web Development Intern

May 2024 – August 2024

- Converted a Figma landing page design into a responsive, production-ready implementation using HTML, CSS, and JavaScript, building a custom image carousel and resolving absolute-positioned elements across device sizes and zoom levels.
- Collaborated remotely with designers and project leadership through iterative feedback and weekly progress updates to refine and deliver web features.

PROJECTS

NEXUS | Python, FastAPI, SQLAlchemy, PostgreSQL, Next.js, React, TypeScript

2026 – present

- Architected and built a full-stack volunteer logistics platform to replace the Google Forms/Sheets workflow used by Science Olympiad directors, with planned deployment at multiple tournaments this year.
- Designed a multi-tenant FastAPI backend with 128 REST endpoints across a 31-table PostgreSQL schema, including scoped role-based permissions allowing volunteers to hold different access across tournaments.
- Engineered a schema-aware form engine that preserves submitted responses across live form edits, a validate-before-write Google Sheets sync pipeline, and JWT/API-key authentication with scoped permissions; backed by 600+ Pytest tests.
- Designed the repository and development workflow for future contributors, using feature branches, pull requests, and issue tracking to support parallel development and maintainable handoffs.

CLUBS AND ORGANIZATIONS

Science Olympiad

Chapter Co-Lead

2026 – present

- Organize the Orange County Regional Tournament serving 80+ teams and coordinate UCI volunteers for tournaments across Southern California.

Vice Captain and Cabinet Officer

2021 – 2025

- Led team operations and mentored members across multiple STEM events, organizing hands-on training and STEM exploration camps to develop technical skills and sustain long-term team participation.

TECHNICAL SKILLS

Languages: Python, SQL, C++, TypeScript, Java, JavaScript, HTML/CSS

Frameworks: FastAPI, Next.js, React, PyQt, Tailwind CSS, Node.js

Developer Tools: Git, GitHub, VS Code, SQLite3, PostgreSQL